

Placebo effects on analgesia related to testosterone and premotor activation

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Using functional magnetic resonance imaging, we tested whether graded placebo conditions could modulate the degree of placebo effect and brain activation patterns in study participants and whether the placebo effect could be influenced by hormones. Each participant was investigated under three conditions: the control (no placebo) condition, the low-placebo condition, and the high-placebo condition (HPC). Activations of the premotor areas, anterior cingulate cortex, and prefrontal cortex were stronger in the HPC compared with those in the control and low placebo conditions. The premotor areas were activated by increased testosterone levels under the HPC. These results suggest that testosterone may affect the brain activation and response to pain during a high-placebo response, with the data supported by brain imaging. *NeuroReport* 22:419–423 © 2011 Wolters Kluwer Health | Lippincott Williams & Wilkins.

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Introduction

The placebo effect describes the outcome after any dummy treatment [1]. In the clinical management of pain, placebo analgesia is the reduction of pain after receiving a substance or undergoing a procedure, where the reduction is not due to the inherent powers of the substance or procedure [2]. In postoperative pain after oral surgery, the placebo effect after an injection of saline solution in full view of the patient has been shown to correspond with the effect of a hidden injection of 8 mg of morphine [3], suggesting a potent placebo effect. Given that pain is a multidimensional subjective experience, the placebo effect may be induced through multiple mechanisms.

Earlier reports have shown that the magnitude of expectation of placebo analgesia is associated with opioid-mediated mechanisms in some cases and with nonopioid-mediated mechanisms in other cases [4]. It is believed that testosterone may modulate opioid analgesia in animals [5]. Low human serum testosterone levels have been reported during psychological and physical stress, as well as in association with the acute stress of surgery [6]. Men with low levels of testosterone tend to be more anxious and irritable than the men with normal

testosterone levels [7]. However, it is not known whether the brain response to pain stimulation could be affected by testosterone.

Therefore, we used functional magnetic resonance imaging (fMRI) to investigate whether the degree of the observed placebo effect and brain activation patterns were changed due to suggestions enabling participants to expect doctors of different caliber and analgesics of different potency. We also investigated the relationship between serum testosterone level and brain response to pain stimulation under control and placebo conditions.

Methods

Fifteen healthy male participants aged 25.33 ± 2.26 years (right-handed) were recruited for participation in this study. The Medical Ethics Committee of Yonsei University Wonju College of Medicine approved this study. All volunteers provided written, informed consent acknowledging that they would experience experimental electrical pain.

Participants were also told that the experiment would identify the effects of medicines with different potencies, prescribed by doctors at different stages of their careers. They were informed that the one potent placebo analgesic was 100 times less potent than the 100 potent placebo analgesic. Furthermore, they were told that the medicines were very short-acting analgesics which would

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take effect immediately after administration, would have a peak effect 10–20 s after administration, and then would expire 10 s later. Participants were not aware that the goal of the experiment was to investigate placebo analgesia.

During the fMRI scanning session, each participant was tested under three conditions: the control condition (no placebo), the low-placebo condition (LPC), and the high-placebo condition (HPC). There were 15 pain stimulation periods during fMRI scanning, each lasting for 15 s, with five pain stimulation periods for each condition (Supplementary Fig. S1, Supplemental digital content 1, <http://links.lww.com/WNR/A115>). Therefore, each participant received 15 pain stimulations (five each in the control condition, the LPC, and the HPC). To reduce habituation, the three conditions were presented in a counterbalanced order. Visual cue was presented in 9-s periods before the pain stimulation was applied. In the control (no placebo) condition, participants were shown a visual cue informing them that no medicines would be injected through the syringe. In the LPC, participants watched a visual cue informing them that one potent analgesic medicine, prescribed by a medical intern having only one year of experience, was being injected through syringe. In the HPC, participants were shown a visual cue informing them that a 100 potent analgesic medicine, prescribed by a pain specialist with a 30-year career, was being injected through syringe. The fMRI scanning session lasted total 15 min and 15 s.

Aside from the infusion of a 0.9% sodium chloride solution into the participants during fMRI scanning, no other medicines or substances were administered under the three conditions. Each participant received the same level of electrical stimulation (2 Hz, 20 mA, duration: 15 s) during fMRI scanning under each of the three conditions. Electrical pain stimulation was provided by a commercial electric stimulation device (SM-920, SamsonMed, Seoul, Korea) and was applied to the dorsal side of the left middle finger using a silver chloride electrode (Hurev Co., Ltd., Wonju, South Korea).

Before fMRI scanning, participants participated in a practice session in which electrical current was surreptitiously reduced from 20 to 16 mA in the LPC and to 12 mA in the HPC. Therefore, participants would confirm the general expectations that the 100 potent analgesics are stronger than the one potent analgesic and that a pain specialist with a 30-year career is better than a medical intern. At the completion of the fMRI scanning session, participants were asked to rate their average perceived pain intensity and anxiety. Ratings were assessed using a Numerical Rating Scale ranging from 0 to 100 (0 = no pain or anxiety; 100 = maximum imaginable pain or anxiety). The ratings of pain and anxiety in the three conditions were analyzed using analysis of variance. An overall *P* value of less than 0.05 was considered significant.

A 20G catheter was placed into the right arm and a venous blood sample was drawn before the fMRI scan to determine the serum testosterone and cortisol levels. Testosterone levels were measured using a γ -counter (COBRA 5010 II, quantum; Packard, USA), with a Coat-A-COUNT total testosterone kit (SIEMENS, Los Angeles, California, USA). The coefficients of variation of testosterone were 5.90%. This method was certificated by College of American Pathologists.

Image acquisition and data analysis

Visual cues were projected onto an LCD screen mounted on top of the head coil. Images were acquired using a 3T MRI scanner (ISOL Technology, Gwangju, Gyeonggi-do, Korea), with a quadrature head coil. Thirty-five slices of fMRI data were acquired using a single-shot gradient echo echo-planar imaging pulse sequence with the following imaging parameters: echo time = 30 ms, repetition time = 3000 ms, flip angle = 80°, matrix size = 64 × 64, field of view = 240 × 240 mm², slice thickness = 4 mm, and voxel size = 3.75 × 3.75 × 4 mm³. Each experimental run produced 305 images, of which the initial five for each scan were discarded.

The imaging analysis was carried out using FSL (FMRIB Software Library, Oxford University, UK). Preprocessing included motion correction using MCFLIRT (FMRIB Software Library, Oxford University, UK) [8], nonbrain removal using BET (FMRIB Software Library) [9], spatial smoothing using a Gaussian kernel with a 5 mm full width at half-maximum, and nonlinear high pass temporal filtering. Functional images were then registered to T1 high-resolution structural images, and finally to a standard brain template (Montreal Neurological Institute 152 stereotaxic template) using FLIRT [8].

In first level analysis, the pain stimulation periods were contrasted with the no pain stimulation periods. In the first and higher-level analyses, activated clusters were reported for *Z* greater than 2.3 and cluster corrected for *P* value of less than 0.05. Then, each participant's mean parameter estimate (PE) was acquired from region of interest (ROI) analysis using Featquery of FMRIB Software Library. To assess the degrees of the responses in pain-related brain areas, statistical analyses among the PEs of the three conditions were compared. The brain areas in which the ROI analysis was performed were the right insula, right prefrontal cortex, premotor area [Brodmann area (BA) 6], and bilateral anterior cingulate cortex (ACC), because they were concurrently activated during the pain stimulation periods of the three conditions in one sample *t*-test (Supplementary Table S1–S3 and Figs S2–S4, Supplemental digital content 1, <http://links.lww.com/WNR/A115>), and the primary somatosensory cortex (S1) and secondary somatosensory cortex (S2), because they are most commonly activated in response to pain stimulation [10]. As the premotor areas

were activated when testosterone levels were increased during pain stimulation periods in the HPC, a simple regression (correlation) analysis was performed to obtain the correlation coefficient between premotor activation and testosterone level (performed using SPM8, Wellcome Department of Cognitive Neurology, London, UK). A volume of interest was selected in the right premotor area [middle frontal cortex, BA 6, (30, -6, 60)], and a 10 mm sphere radius was used. The correlation coefficient between the eigenvariate value (premotor activation) and testosterone level was acquired.

The correlation coefficients between the PEs, as well as between the PEs and the behavioral ratings, were calculated using a correlation analysis. An overall P value of less than 0.05 was considered significant.

Results

Pain ratings [control condition (59.83 ± 12.15), LPC (54.00 ± 13.26), HPC (43.67 ± 13.82)] were significantly different among the three conditions [$F(2,42) = 5.864$, $P = 0.006$]. Pain ratings were significantly higher in the control condition and LPC than in the HPC ($P = 0.002$ and 0.036 , respectively). Anxiety ratings [control condition (43.83 ± 18.15), LPC (36.50 ± 16.66), HPC (27.27 ± 13.60)] were also significantly different among the three conditions [$F(2,42) = 3.917$, $P = 0.028$]. Anxiety ratings were significantly higher in the control condition than in the HPC ($P = 0.008$). The overall mean testosterone level was $5.05 \pm 1.20 \mu\text{g/dl}$ (mean $\pm$ standard deviation) (normal range: $2.62\text{--}15.93 \mu\text{g/dl}$). These results indicate that participants believed that the HPC was more effective than the LPC.

In the ROI analysis (Supplementary Table S4, Supplemental digital content 1, <http://links.lww.com/WNR/A115>), the right middle frontal gyrus (MFG, BA 6, premotor area, $P = 0.038$), right superior frontal gyrus (SFG, BA 6, premotor area, $P = 0.014$), left ACC (BA 32, $P = 0.023$), right SFG (BA 8, $P = 0.026$), and right MFG (BA 46, $P = 0.032$) were all more strongly activated in the HPC than in the control condition. The left ACC (BA 32, $P = 0.035$) was more strongly activated in the HPC than in the LPC. The right MFG (BA 46, $P = 0.015$) was more strongly activated in the LPC than in the control condition. The S1 (BA 1, $P = 0.044$) was more strongly activated in the control condition than in the HPC.

The bilateral premotor areas {[right MFG (30, 0, 62): voxels (372), Z score (3.61)], [left MFG (-28, -8, 48): voxels (680), Z score (3.24)]} were activated in the HPC during pain stimulation when testosterone levels were increased (Fig. 1a and Supplementary Table S5, Supplemental digital content 1, <http://links.lww.com/WNR/A115>), but not in the control condition or LPC.

The correlation coefficient (r) between the testosterone level and the brain activation (eigenvariate values) in the

right MFG (BA 6) was 0.836 ($P < 0.001$) during the pain stimulation periods of the HPC (Fig. 1b). The r between brain activation (PEs) of the right ACC (BA 32) and brain activation (PEs) of the right MFG (premotor area, BA 6) was 0.537 ($P = 0.039$, Fig. 1c) during the pain stimulation periods in the HPC.

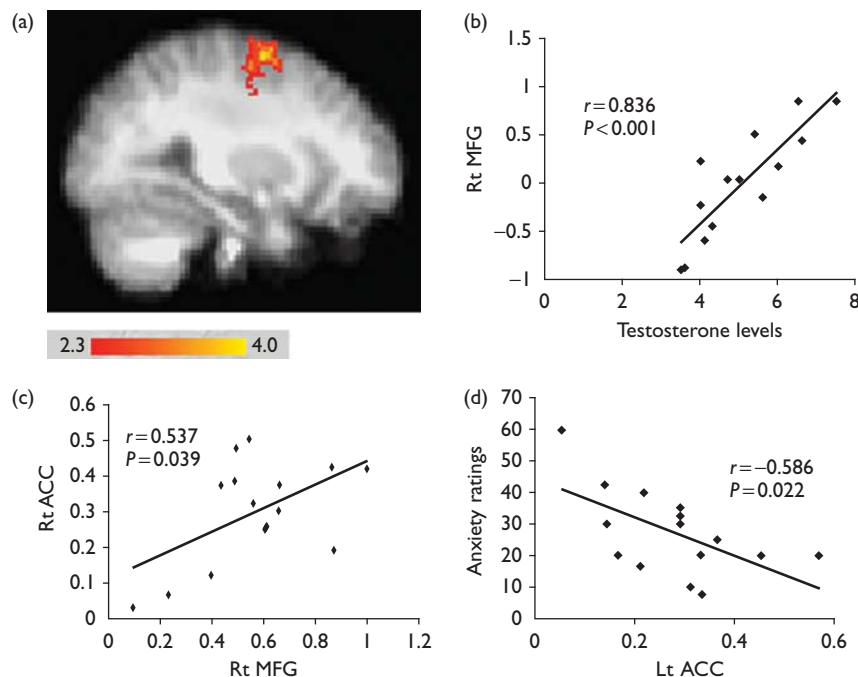
In addition, correlations were found between brain activation (PEs) and behavioral data, as shown in Fig. 1d: the r between the PE in the right MFG (BA 6) and the pain rating in the control condition = -0.523 ($P = 0.046$); the r between the PE in the right SFG (BA 6) and the pain rating in the control condition = -0.588 ($P = 0.021$); the r between the PE in the right SFG (BA 8) and the pain rating in the LPC = -0.546 ($P = 0.035$); the r between the PE in the left ACC (BA 32) and the pain rating in the control condition = -0.545 ($P = 0.036$); the r between the PE in the left ACC (BA 32) and the anxiety rating in the HPC = -0.586 ($P = 0.022$); the r between the PE in the right ACC (BA 32) and the anxiety rating in the HPC = -0.636 ($P = 0.011$).

Discussion

Relationships among testosterone, placebo, and brain activations in the premotor areas

In the ROI analysis, the premotor areas were more strongly activated in the HPC than in the control condition. An earlier study showed that the tendency for increased neuronal activity after cues predicting a large reward became progressively stronger in more posterior areas in the lateral section of the frontal lobe (dorsolateral prefrontal cortex < frontal eye field < transitional zone caudal to the frontal eye field < premotor cortex) [11]. It is also known that placebos can have reward responses [12,13].

In this study, the relationship between testosterone level and premotor activation during pain stimulation periods in the HPC was strongly supported by our earlier study on female participants ($n = 18$, aged 23.11 ± 1.91 years) [14]. In that study, serum testosterone levels were negatively correlated with pain ratings during the follicular phase of the menstrual cycle ($r = -0.612$, $P = 0.007$). Activation of the premotor area during pain stimulation was positively correlated with the testosterone level during the follicular phase ($r = 0.819$, $P = 0.002$). Activation of the right MFG (premotor area) during pain stimulation was negatively correlated with the pain rating during the follicular phase ($r = -0.747$, $P = 0.01$). Activation of the bilateral premotor areas during pain stimulation were negatively correlated with pain ratings during the luteal phase of the menstrual cycle [right SFG ($r = -0.792$, $P = 0.004$), left MFG ($r = -0.804$, $P = 0.003$)]. This data about female participants suggest that the testosterone level might be involved in premotor activation during a response to pain stimulation, and that premotor activations might decrease pain perception in women.

Fig. 1

(a) Premotor areas [right middle frontal gyrus (Rt MFG) (30, 0, 62): voxels (372), Z score (3.61)] activated during pain stimulation periods of the high-placebo condition (HPC) when testosterone level was increased. Activated clusters were reported for Z greater than 2.3 and cluster corrected *P* value of less than 0.05. (b) Positive correlation between testosterone level (micrograms per deciliter) and brain activation (eigenvariate value) in the Rt MFG (premotor area, BA 6) during pain stimulation periods of the HPC. *r*, correlation coefficient. (c) Positive correlation between the brain activation (parameter estimate = PE) in the right ACC (Rt ACC, BA32) and the brain activation (PE) in the right middle frontal gyrus (Rt MFG, premotor area, BA 6) during pain stimulation periods of the high placebo condition. (d) Negative correlation between the anxiety rating and the brain activation (PE) in the left ACC (Lt ACC, BA32) during pain stimulation periods of the high placebo condition.

In this study, we found that, in the HPC, the premotor areas were more strongly activated when the placebo effect and testosterone level increased, as well as when the participant perceived less pain during pain stimulation periods. The premotor areas were more strongly activated when the pain ratings of the participants decreased during pain stimulation in the control condition. In the behavioral pain ratings, participants perceived less pain in the HPC than in the control condition and LPC. In addition, in the ROI analysis, premotor areas were more strongly activated in the HPC than in the control condition. Finally, the premotor areas were more strongly activated in participants with higher serum testosterone levels during the pain stimulation periods of the HPC than in those with lower levels. Therefore, these premotor areas might be activated to prepare motor and defense responses to pain stimulation and to decrease pain perception in the HPC.

Furthermore, because there was a positive correlation between premotor activation and right ACC activation in the HPC (Fig. 1c), participants with higher serum testosterone levels should have experienced activation in the premotor areas and ACC. As increased activity in the ACC is associated with both placebo and opioid

analgesia [15–19], premotor activation in the HPC might indicate effective placebo analgesia.

Anterior cingulate cortex and the placebo effect

The left ACC (BA 32) was more strongly activated in the HPC than in the control condition or LPC. In addition, the left ACC was more strongly activated when pain ratings decreased during pain stimulation in the control condition. The bilateral ACC were more strongly activated when anxiety ratings decreased during pain stimulation in the HPC. These results indicate that participants with lower pain and anxiety ratings had greater ACC (BA 32) activation compared with that of participants with higher ratings, to cope with their suffering. As both placebo and opioid analgesia are associated with increased activity in the ACC [15–19], the placebo effect in this study might be related to the endogenous opioid system that is believed to be activated by a higher testosterone level. The above data suggest that the ACC might be involved in the placebo response.

Prefrontal cortex and the placebo effect

It has been reported that placebo analgesia was completely blocked when the prefrontal cortex was suppressed

by noninvasive, low frequency repetitive transcranial magnetic stimulation [20]. The prefrontal cortex is known to exert active control over pain perception, the modulation of pain, and in limiting the extent of suffering in a top-down manner [21]. The right SFG (BA 8) was more strongly activated when pain ratings decreased during pain stimulation periods in the LPC. An earlier study showed that activation in the bilateral anterolateral prefrontal (BA 46) cortices was negatively correlated with pain intensity rating [22]. In our behavioral data, pain ratings decreased more in the placebo conditions than they did in the control condition. The right MFG (BA 46) was more strongly activated in the HPC and LPC than in the control condition. Therefore, activations of the prefrontal cortices might lead to greater pain relief in the placebo conditions than in the control condition.

S1 and pain

It has been shown that the S1 response to pain stimulation exhibits a steady increase in activation with applied stimulus intensity [23]. Activation of the S1 (BA1) was stronger in the control condition than in the HPC. Therefore, participants might perceive less pain in the HPC than in the control condition.

In this study, a significant correlation between brain activation and cortisol level was not found. However, there are some reports indicating a significant correlation between stress and cortisol level. Further study is necessary to clarify the underlying mechanism of this study.

Conclusion

These results indicate that the expectation of a HPC can provide greater pain relief compared with expectation of a LPC. These results suggest that testosterone may affect the brain activation and response to pain during the HPC.

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